

Alex Hasenbein

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EDUCATION

The Pennsylvania State University

B.S. Computer Science, B.S. Mathematics | GPA: 3.48

University Park, PA

May 2028

Awards and Honors:

- Behrend Honors Program, Deans List
- 1st Place in AI Agents and LLMs @ PSU ACM x Behrend Hackathon
- VP of Pickleball club, Treasurer of Entrepreneurship Club

WORK EXPERIENCE

Flex

Austin, TX

Software Engineering and Machine Learning Intern (Process Engineering Team)

May 2026 - Present

- Built and deployed a full ML pipeline in Python (**LightGBM, pandas, scikit-learn**) processing **1M+ records** to forecast throughput, delivering 90% accuracy and sub-30-second precision on time predictions.
- Engineered features from SQL Server views and optimized complex queries across production datasets of **4M+ rows**, implementing parallelized chunking to eliminate tempdb exhaustion.
- Designed web-based dashboards in C# (ASP.NET) with dynamic charts, tables, and filters, delivering real-time analytics for the process engineering team. Lead two HQ wide projects- overseeing version control, monitoring prs handled & roadmaps.
- Created data analysis apps, built in Python using Plotly, Pandas, and Pydoc **that reduced team wide analysis time by 10%**.

The Pennsylvania State University

Erie, PA

Undergraduate Researcher, Honors Option

Jan 2025 – May 2025

- Built a hyperparameter optimization framework for neural networks predicting quadratic roots, benchmarking Bayesian optimization (Optuna), random search, and multi-objective pareto search to quantify accuracy-efficiency tradeoffs.
- Built a data-pipeline and full evaluation suite (MAE, RMSE, R², per class-error, inference speed) with interactive dashboards for hyperparameter analysis, trial exploration, ensuring reproducible, interpretable results.

PROJECTS AND HACKTHONS

DevCord – Multi Agent AI Orchestration Platform | 1st Place, AI Agents & LLMs : Behrend Hackathon

TypeScript, Discord.js, and OpenRouter

Won 1st place in the most competitive category, **building and shipping solo in 10 hours** a multi-agent platform that turns natural-language Discord messages into fully managed software execution pipelines

- Architected specialized agents (Manager, Dev, QA, Command) with scoped roles and task handoff in strict TypeScript using Discord.js and OpenRouter, with modular model routing to swap LLMs per agent without touching core orchestration logic
- Implemented autonomous project kickoff, thread-based task decomposition, persistent run/state tracking with pause/resume/cancel controls, plus voice-first Q&A via local Whisper STT + TTS in Discord voice channels

Particles Interlinked

Next.js, React, Three.js, Node.js

- Developed a complex real-time particle simulation system using Three.js with WebGL rendering, applying physics concepts such as particle interactions, gravitational attraction, and collision detection implemented efficiently in JavaScript.
- Integrated performance monitoring and leveraged WebGL for hardware-accelerated graphics rendering, demonstrating strong skills in performance tuning and UI/UX for interactive simulations, **increasing fps by 22% consistently**.
- Switched from a 2d to a 3d library to make graphics more dynamic while optimizing to allow **faster rendering than the 2d**.

BowieForum

React, Mongo DB, SQL

- Designed and deployed a student forum platform with 150+ users. Utilized optimized SQL queries to reduce server load.
- Used pagination to efficiently load posts to reduce strain both client side and server side while keeping low latency.

CERTIFICATIONS, SKILLS, TECHNOLOGIES & INTERESTS

- **Programming Languages:** Java, C++, C#, JavaScript, TypeScript, Python, SQL, Rust, VB.NET
- **Frameworks:** React, Next.js, Node.js, Spring, ASP.NET, FastAPI, Flask, LightGBM, pandas, scikit-learn, PlotLy
- **Tools & Technologies:** Git, Docker, MongoDB, SQL Servers, Node.js, OpenRouter, CI/CD Pipelines
- **Focus Areas:** Full-Stack Development, Machine Learning & AI, Data Science, Backend Engineering, Multi-Agent Systems